A simple RAG demo using OpenAlex titles and abstracts

Code

Exploring Conservation Reserve Program literature with ragnar and ellmer

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Purpose

This short demonstration shows how we can build a simple retrieval-augmented generation (RAG) workflow from a large JSON file containing bibliographic records.

The example uses records from OpenAlex relating to the **Conservation Reserve Program**. Each record contains a title, abstract, DOI, publication year and journal. The aim is not to create a definitive evidence synthesis, but to demonstrate how RAG can help us explore a large body of literature.

Important caveat: this workflow uses **titles and abstracts only**. The chatbot should therefore be treated as a literature exploration and evidence mapping assistant, not as a full systematic review or full-text evidence synthesis tool.

Setup

You will need the following R packages.

```
# install.packages(c(
#   "jsonlite",
#   "dplyr",
#   "stringr",
#   "ragnar",
#   "ellmer",
#   "vctrs"
# ))

library(jsonlite)
library(dplyr)
library(stringr)
library(ragnar)
library(ellmer)
library(vctrs)
```

You also need an OpenAI API key available to R. One common option is to set it in `.Renviron`:

```
OPENAI_API_KEY="your-key-here"
```

Then restart R.

Load the JSON file

The JSON file is expected to contain a top-level object called `records`.

```r
raw <- fromJSON("crp_openalex_enhanced.json")

records <- raw$records

glimpse(records)
```

Prepare the records

We keep only the fields needed for retrieval and citation.

```r
docs <- records |>
  transmute(
    id = as.character(id),
    doi = as.character(doi),
    title = as.character(title),
    abstract = as.character(abstract),
    year = as.integer(publication_year),
    journal = as.character(journal)
  ) |>
  filter(
    !is.na(title),
    !is.na(abstract),
    abstract != ""
  )

nrow(docs)
```

Helper function for cleaning text

Some OpenAlex records contain unusual spacing, encoding artefacts, or hidden control characters. This helper function makes the text safer to send to the embedding model.

```r
#' Clean text before embedding
#'
#' Converts text to UTF-8, removes non-printing characters, removes control
#' characters, and squishes repeated whitespace.
#'
#' @param x A character vector.
#'
#' @return A cleaned character vector.
clean_text <- function(x) {
  x |>
    as.character() |>
    iconv(from = "", to = "UTF-8", sub = " ") |>
    str_replace_all("[^[:print:]\n\r\t]", " ") |>
    str_replace_all("[[:cntrl:]]", " ") |>
```

```
    str_squish()
}
```

Create chunks for RAG

For this demonstration, each title and abstract becomes one retrieval unit.

For full-text documents we would usually chunk the text more carefully. For abstracts, keeping one record as one chunk is usually sufficient.

```
chunks_rag <- docs |>
  transmute(
    id = id,
    doi = if_else(is.na(doi), "", doi),
    title = clean_text(title),
    year = year,
    journal = if_else(is.na(journal), "", clean_text(journal)),
    abstract = clean_text(abstract),
    text = str_glue("{title}\n\n{abstract}")
  ) |>
  filter(
    !is.na(text),
    text != "",
    nchar(text) > 20
  ) |>
  mutate(
    # Keeps the embedding requests manageable.
    # This is usually enough for titles and abstracts.
    text = str_sub(text, 1, 6000)
  )

glimpse(chunks_rag)
```

Create the ragnar store

This creates a DuckDB-backed vector store. We use OpenAI's `text-embedding-3-small` model for embeddings.

```
store <- ragnar_store_create(
  "crp_abstracts.duckdb",
  embed = \(x) embed_openai(x, model = "text-embedding-3-small"),
  extra_cols = vctrs::vec_ptype(chunks_rag),
  version = 1,
  overwrite = TRUE
)
```

Insert records in small batches

With thousands of abstracts, inserting everything at once can create embedding requests that are too large. Batching makes the process more reliable.

```r
#' Insert chunks into a ragnar store in batches
#'
#' This avoids sending too many texts to the embedding API in one request.
#'
#' @param store A ragnar store object created with `ragnar_store_create()`.
#' @param chunks A data frame containing a `text` column and metadata columns.
#' @param batch_size Number of records to insert per batch.
#'
#' @return Invisibly returns `NULL`.
insert_in_batches <- function(store, chunks, batch_size = 50) {
  for (i in seq(1, nrow(chunks), by = batch_size)) {
    j <- min(i + batch_size - 1, nrow(chunks))

    message("Inserting rows ", i, " to ", j)

    ragnar_store_insert(
      store,
      chunks[i:j, ]
    )
  }

  invisible(NULL)
}

insert_in_batches(store, chunks_rag, batch_size = 50)
```

Build the retrieval index

```r
ragnar_store_build_index(store)
```

Test retrieval directly

Before adding a chatbot layer, it is useful to check whether retrieval is returning sensible records.

```r
ragnar_retrieve(
  store,
  "What evidence is there that the Conservation Reserve Program reduces nitrate losses?",
  top_k = 10
) |>
  select(title, year, journal, doi)
```

```
# A tibble: 20 � 5
   title                                                                          year journal doi
text
   <chr>                                                                         <int> <chr>   <chr>
<chr>
```

```
 1 Groundwater Nitrate Concentration Reductions in a Riparian Buffer En.  2014 "JAWRA. "htt.
"Gro.
 2 Nitrate Removal of Riparian Buffers Established Through the NC Conse.  2006 "2006 . "htt.
"Nit.
 3 Restoring Wetlands for Nitrate Removal from Subsurface Drainage: Res.  2013 ""      "htt.
"Res.
 4 Nitrogen Dynamics Affected by Management Practices in Croplands Tran.  2014 "Agron. "htt.
"Nit.
 5 Ecological Benefits of the Conservation Reserve Program                1993 "Conse. "htt.
"Eco.
 6 Nitrate Losses through Subsurface Tile Drainage in Conservation Rese.  1997 "Journ. "htt.
"Nit.
 7 Hydrological impacts of the conservation reserve program-a mini revi.  2024 "Front. "htt.
"Hyd.
 8 Conservation Reserve Program effects on soil quality indicators        1999 "Journ. "htt.
"Con.
 9 Groundwater nitrate reduction processes in a riparian buffer enrolle.  2010 "2010 . "htt.
"Gro.
10 Water Quality Benefits from the Conservation Reserve Program           1989 ""      ""
"Wat.
11 JAWRA: Fifty-Five Years of Sustained Contributions for Improved Wate.  2019 "JAWRA. "htt.
"JAW.
12 �tude sur la mise en ouvre d'outils d'�cofiscalit� au service de la .  2023 ""      "htt.
"�tu.
13 �tude sur la mise en ouvre d'outils d'�cofiscalit� au service de la .  2023 ""      "htt.
"�tu.
14 Management plan and conservation strategies for sage grouse in Monta.  2005 ""      "htt.
"Man.
15 Adaptive management in native grasslands managed by the U.S. Fish an.  2018 "Antar. "htt.
"Ada.
16 ECONOMIC AND STOCHASTIC EFFICIENCY COMPARISON OF EXPERIMENTAL TILLAG.  2011 "Exper. "htt.
"ECO.
17 SWITCHGRASS POTENTIAL ON RECLAIMED SURFACE MINES FOR BIOFUEL PRODUCT.  2012 "Recla. "htt.
"SWI.
18 Switchgrass potential on reclaimed surface mines for biofuel product.  2012 ""      "htt.
"Swi.
19 Management and research applications of long-range surveillance rada.  2008 "Fact . "htt.
"Man.
20 Hydrostratigraphic interpretation of test-hole and borehole geophysi.  2014 "Antar. "htt.
"Hyd.
```

Example retrieval questions

```
queries <- c(
  "biodiversity benefits of conservation reserve program",
  "water quality nitrate runoff tile drainage",
  "land use change slippage effects",
  "economic evaluation conservation reserve program",
  "soil organic carbon storage conservation reserve program",
  "grassland bird responses conservation reserve program"
)
```

```
retrieval_examples <- lapply(queries, \(q) {
  ragnar_retrieve(store, q, top_k = 5) |>
    select(title, year, journal, doi)
})

names(retrieval_examples) <- queries

retrieval_examples
```

Create a chatbot

The system prompt is deliberately cautious. The model is told that it is using titles and abstracts, not full texts.

```
chat <- chat_openai(
  model = "gpt-4.1",
  system_prompt = "
You answer questions using retrieved titles and abstracts.

Important constraints:
- You are using titles and abstracts, not full texts.
- Do not overstate the strength of evidence.
- Cite the title, year, journal and DOI where possible.
- If the retrieved evidence is weak, mixed, or irrelevant, say so.
- If the question cannot be answered from the retrieved abstracts, say so clearly.
"
)

ragnar_register_tool_retrieve(chat, store)
```

Example chatbot responses

Example 1: Broad mapping question

```
chat$chat(
  "What are the main themes in this literature on the Conservation Reserve Program?"
)
```

```
The literature on the Conservation Reserve Program (CRP) centers on several main themes:

1. **Program Structure and Purpose**: The CRP, established in 1985, is designed to retire
highly
erodible or environmentally sensitive cropland, typically through 10-year contracts. The goal
is
to conserve soil and water resources, and participants receive rental payments, cost-sharing,
and
technical assistance to establish approved plantings (see "Conservation Reserve Program:
Status
and Policy Issues," 1997; "Conservation Reserve Program: Status and Current Issues," 2005).

2. **Land Retirements and Environmental Benefits**: The CRP is the largest federal land
```

```
retirement program, reducing active farming on marginal lands to facilitate ecosystem recovery
and environmental benefits such as improved water quality, reduced soil erosion, and enhanced
wildlife habitat (see above titles).

3. **Biodiversity and Wildlife**: The program influences wildlife, especially grassland birds
and
pollinators. For example, integrating prairie strips in crop fields can increase pollinator
and
bird abundance, as well as affect nutrient and sediment runoff ("Assessing the effects of
embedding prairie strips within row crop fields on soil hydraulic properties and soil health,"
2022).

4. **Soil Health and Water Quality**: Studies assess how conservation measures like prairie
strips impact soil hydraulic properties, wet-aggregate stability, runoff, and overall soil
health. These effects, however, may vary across site conditions and soil history.

5. **Policy and Administration**: The CRP is administered by the USDA's Farm Service Agency,
with
technical support from the Natural Resources Conservation Service. Policy issues include
enrollment caps, eligibility, and the design of incentive structures.

6. **Management of Native and Exotic Grasslands**: Research compares bird and plant diversity,
as
well as ecosystem processes, in native versus exotic grasslands-much of which overlaps with
CRP
incentive areas and management strategies for retired farmland ("Grassland birds wintering at
U.S. Navy facilities in southern Texas," 2010).

7. **Societal Perceptions and Adoption**: Perception among farmers and the broader public
about
the value and implementation of CRP and related conservation practices like prairie strips can
affect program participation and success.

In summary, themes cover program operations and policy, environmental and biodiversity
outcomes,
impacts on soil and water, comparison of native vs. restored plant communities, and the social
dynamics influencing program adoption.

**Key Sources**:
- "Conservation Reserve Program: Status and Policy Issues," 1997.
- "Conservation Reserve Program: Status and Current Issues," 2005.
- "Assessing the effects of embedding prairie strips within row crop fields on soil hydraulic
properties and soil health," 2022. https://doi.org/10.31274/td-20240329-92
- "Grassland birds wintering at U.S. Navy facilities in southern Texas," 2010.
https://doi.org/10.3133/ofr20101115

(The evidence for some themes, like biodiversity and soil health impacts, is robust; however,
more specific studies may exist if detailed subtopics are of interest.)
```

Example 2: Specific environmental outcome

```
chat$chat(
  "What evidence is there that the Conservation Reserve Program improves water quality or redu
```

```
)
```

There is direct evidence that the Conservation Reserve Program (CRP), particularly through its

quality:

1. **Groundwater Nitrate Reduction**:
   - A study of a 46-meter-wide riparian buffer enrolled in the North Carolina Conservation Reserve Enhancement Program (NC CREP, a state-level version of CRP) found that groundwater nitrate-nitrogen concentrations were reduced by 76-92% across the buffer. This reduction was observed in all monitored sections and was attributed to a combination of denitrification, plant uptake, and dilution processes. Chloride reductions (48-65%) helped demonstrate that dilution was a contributing factor, but biological processes played a key role as well ("Groundwater Nitrate Concentration Reductions in a Riparian Buffer Enrolled in the NC Conservation Reserve Enhancement Program," JAWRA Journal of the American Water Resources Association, 2014, https://doi.org/10.1111/jawr.12209).

2. **Effectiveness of Forested Riparian Buffers**:
   - Two forested riparian buffers established under the NC CREP were monitored. Groundwater nitrate levels were dramatically reduced as water moved from the agricultural field edge to the stream edge (from averages of 8.1 mg/L and 4.1 mg/L down to 1.3 mg/L and 1.6 mg/L in shallow wells). This study concluded that these buffers significantly reduced nitrate loads in groundwater before reaching streams, likely limiting further increases in stream nitrate concentrations ("Nitrate Removal of Riparian Buffers Established Through the NC Conservation Reserve Enhancement Program," 2006, https://doi.org/10.13031/2013.21438).

3. **Longitudinal Monitoring**:
   - Over several years, buffers of multiple vegetation zones (hardwood, pine, grass) enrolled in the NC CREP were monitored for nitrate in groundwater and adjacent streams. The studies consistently found significant reductions in nitrate concentrations in groundwater as it passed through the buffer, reinforcing the buffer's water quality benefits ("Groundwater nitrate reduction processes in a riparian buffer enrolled in the NC Conservation Reserve Enhancement Program," 2010, https://doi.org/10.13031/2013.29893).

Summary: Riparian buffers established via the Conservation Reserve Program or similar programs have been shown to reduce nitrate concentrations in groundwater by significant margins (often 75% or more). These reductions help prevent agricultural nitrate from reaching streams and water bodies, improving overall water quality. The main mechanisms are plant uptake, microbial denitrification, and dilution. The effect is most pronounced in groundwater; impact on stream concentrations depends on specific site conditions.

References:
- Deal, N.E., et al. (2014). Groundwater Nitrate Concentration Reductions in a Riparian Buffer Enrolled in the NC Conservation Reserve Enhancement Program. *JAWRA J Am Water Resour Assoc*,

50(3), 653-664. https://doi.org/10.1111/jawr.12209
- Evans, R.O., et al. (2006). Nitrate Removal of Riparian Buffers Established Through the NC Conservation Reserve Enhancement Program. https://doi.org/10.13031/2013.21438
- Deal, N.E., et al. (2010). Groundwater nitrate reduction processes in a riparian buffer enrolled in the NC Conservation Reserve Enhancement Program.
https://doi.org/10.13031/2013.29893

These findings are robust for riparian (streamside) buffer practices funded under CRP/CREP, particularly in the context of North Carolina, but the principle is generalizable to other regions using similar conservation buffers.

Example 3: Biodiversity question

```
chat$chat(
  "What does this literature suggest about the effects of the Conservation Reserve Program on
)
```

The literature indicates that the Conservation Reserve Program (CRP) generally has positive effects on grassland birds:

1. **Habitat Creation and Bird Diversity**
CRP fields, through land retirement and restoration of native or perennial vegetation, provide crucial habitat for grassland birds-many of which have declined due to conversion of grassland to
croplands. The presence of CRP lands has been associated with increased abundance and diversity
of grassland birds compared to intensively farmed or exotic grass-dominated lands.

2. **Comparison of Native and Exotic Grasslands**
Studies comparing native and exotic (non-native) grasslands, often in the context of CRP habitats, found native grasslands consistently support greater bird species richness than exotic
grasslands. However, some exotic grasslands enrolled in CRP can have higher total numbers of certain bird species in some years ("Grassland birds wintering at U.S. Navy facilities in southern Texas," Antarctica A Keystone in a Changing World, 2010,
https://doi.org/10.3133/ofr20101115).

3. **Conservation of At-risk or Declining Species**
CRP lands are important for supporting populations of several declining grassland bird species,
including species of conservation concern such as Le Conte's Sparrow, Grasshopper Sparrow, and Loggerhead Shrike. Seven species observed in CRP-associated studies were listed as of conservation concern, and most common birds detected were grassland species (same source above).

4. **Vegetation Complexity and Bird Use**
Species richness and bird abundance in CRP lands are related to the type and complexity of vegetation cover. Native vegetation generally supports a higher diversity of birds, while management practices (e.g., mowing, burning, rest) also affect which bird species are present.

5. **Seasonal Use and Migratory Birds**
CRP fields are not only breeding habitat but are also important for overwintering and

```
migratory
grassland birds, offering food sources and shelter during critical times ("Grassland birds
wintering at U.S. Navy facilities in southern Texas," 2010).

**Summary:**
The evidence from CRP-related studies is robust in demonstrating that the program provides
essential habitat and supports higher diversity and abundance of grassland birds, including
many
at-risk species. Outcomes tend to be most favorable in CRP fields restored to native
vegetation,
but even managed exotic grasslands under CRP offer benefits over conventional agricultural
fields.

**Key Reference:**
- "Grassland birds wintering at U.S. Navy facilities in southern Texas," Antarctica A Keystone
in
a Changing World, 2010. https://doi.org/10.3133/ofr20101115

If you need evidence for particular bird species or want more detail on management practices,
I
can search for additional targeted studies.
```

Example 4: Evidence gaps

```
chat$chat(
  "Based on the retrieved abstracts, what appear to be important evidence gaps or under-studie
)
```

```
Based on the retrieved abstracts, several important evidence gaps and under-studied questions
in
the Conservation Reserve Program (CRP) literature are apparent:

### 1. **Long-term Water Quality Outcomes for Streams**
- While several studies show CRP-associated riparian buffers significantly reduce nitrate in
groundwater before it enters streams, results for streamwater nitrate concentrations are less
clear. One study even notes that "both buffered streams showed increased levels in nitrate
concentration downstream of the buffers," suggesting that groundwater improvement may not
always
translate to immediate stream improvements (Evans et al., 2006).
- **Gap:** There is a need for more long-term, watershed-scale studies documenting how and
when
groundwater nitrate reductions from CRP practices result in measurable improvements in stream
water quality.

### 2. **Causal Mechanisms of Nitrate Reduction**
- The studies found that nitrate reductions in groundwater could be due to a mix of
denitrification, plant uptake, and dilution, with the precise contribution of each process
unclear.
- **Gap:** More research is needed to quantify the relative importance of biotic (e.g.,
denitrification, plant uptake) versus abiotic (e.g., dilution) processes in different settings
to
better predict performance and assign nitrogen removal credits.
```

3. **Impacts on Soil Health and Productivity**
- Recent work on prairie strips in croplands notes that effects on soil health are "not definitively greater in prairie strips than row crops across sites at the current establishment stage." Most marked improvements are in wet-aggregate stability, but broader, longer-term impacts on overall productivity and nutrient cycling remain uncertain ("Assessing the effects of embedding prairie strips within row crop fields...", 2022).
- **Gap:** More studies spanning longer timeframes, different soil types, and climates are needed to fully understand the impacts of CRP practices on soil health and agricultural productivity.

4. **Wildlife Responses-Beyond Birds**
- While there is substantial information on grassland birds, less is known about the response of other wildlife (e.g., mammals, insects, reptiles) to CRP practices. Even within bird studies, most focus on North American breeding birds, with wintering birds ("Grassland birds wintering at U.S. Navy facilities in southern Texas," 2010) and migrating species less studied.
- **Gap:** Comprehensive assessments of the effects of CRP on a wider spectrum of wildlife and across seasons are needed.

5. **Relative Effectiveness of Native vs. Exotic Vegetation**
- Research consistently shows higher bird species richness in native grasslands compared to exotic ones, but there are still years and situations where exotic grasslands support higher bird abundance.
- **Gap:** There is a need for nuanced, species- and context-specific comparisons to determine which wildlife and ecosystem services benefit most from native versus exotic vegetation types under CRP.

6. **Socioeconomic and Adoption Factors**
- One abstract notes the "favorable perception in farming and nonfarming populations" for prairie strips but does not delve deeply into adoption rates, barriers, or economic returns for farmers.
- **Gap:** More detailed, empirical studies on economic outcomes, management trade-offs, and farmer decision-making could help optimize enrollment and design of CRP practices.

7. **Landscape and Management Context**
- Existing studies often focus on individual sites or small-scale setups. There is limited discussion on how landscape context, adjacent land use, and specific management practices (e.g., grazing, burning, mowing) interact with CRP outcomes.
- **Gap:** More landscape-scale, multi-factor studies are needed to inform best practices for CRP implementation and adaptive management.

In summary:
While the CRP literature shows strong benefits for water quality (mainly in groundwater), grassland birds, and potentially soil health, it is often site-specific and outcome-specific.

```
There are clear evidence gaps related to causal mechanisms, multi-scale (especially stream and
landscape scale) impacts, effects on a wider array of wildlife, native versus exotic
vegetation
benefits, socioeconomic factors, and best management practices. These represent important
areas
for future research to maximize the effectiveness and resilience of CRP and similar
conservation
programs.
```

Workflow:

1. We start with a large bibliographic dataset that is too large to inspect manually.
2. We convert each title and abstract into a searchable text unit.
3. We embed those text units, meaning that records can be retrieved by semantic similarity rather than exact keyword matching.
4. We first test retrieval directly, which is important for quality control.
5. We then connect the retrieval system to a chatbot.
6. The chatbot is instructed to answer cautiously because it only has access to titles and abstracts.

The key principle is that the language model is not being asked to answer from memory. It is being asked to answer using retrieved records from a defined corpus.

Important limitations

This should be framed as a **literature exploration tool**, not a replacement for systematic review judgement.

Limitations include:

- The chatbot only sees titles and abstracts.
- Important methodological details may be absent from abstracts.
- Retrieval may miss relevant records if the query is poorly phrased.
- The language model may still over-generalise if not prompted carefully.
- The results should be checked against the retrieved records.
- This is useful for scoping, mapping, question refinement and screening support, but not for final evidence claims.

Possible next steps

Useful extensions could include:

- adding topic codes or evidence map categories;
- clustering abstracts by theme;
- developing a Shiny (interactive html) interface;
- adding full texts where licences and access allow.